



***In silico* toxicity studies of traditional Chinese herbal medicine: A mini review**

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Abstract

With the availability of public databases that store compound-target/compound-toxicity information, and Traditional Chinese medicine (TCM) databases, *in silico* approaches are used in toxicity studies of TCM herbal medicine. Here, three *in silico* approaches for toxicity studies were reviewed, which include machine learning, network toxicology and molecular docking. For each method, its application and implementation e.g., single classifier vs. multiple classifier, single compound vs. multiple compounds, validation vs. screening, were explored. While these methods provide data-driven toxicity prediction that is validated *in vitro* and/or *in vivo*, it is still limited to single compound analysis. In addition, these methods are limited to several types of toxicity, with hepatotoxicity being the most dominant. Future studies involving the testing of combination of compounds on the front end i.e., to generate data for *in silico* modeling, and back end i.e., validate findings from prediction models will advance the *in silico* toxicity modeling of TCM compounds.

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Traditional Chinese medicine, Network toxicology, Machine learning, Molecular docking, Herbal medicine, *In silico* toxicity studies.

Introduction

The use of traditional Chinese medicine (TCM) has steadily increased every year. In China, TCM accounts for

40% of its pharmaceutical market [1] but unlike Western medicine, TCM herbal medicine is usually administered in the form of a decoction of more than one herbal plant, where each constituent of the plant works synergistically. This inherently produces multiple effects such as anti-oxidant and anti-inflammatory. While this strategy exemplifies the more accepted polypharmacology paradigm [2], these herbal medicines also inevitably produce several side effects, although it is possible that the combination of herbs may neutralise or counteract the side effects. Unlike Western medicine, TCM herbal medicines are not subjected to intense scrutiny on their efficacy and safety as it is still considered Complementary Medicine. Every year, it is estimated that 230,000 TCM-related adverse effects were reported [3]. Furthermore, herbal products were ranked second as the source of drug-induced liver injury in the United States [4]. Although significant efforts have been made to address these, the safety profile of TCM herbal medicine is still lacking despite the steady increase in its use.

One approach that can be considered in establishing the safety profile of TCM herbal medicine is the use of *in silico* methods. This is made possible by the availability of public databases that store compound-target/compound-toxicity information that can be used in both mechanisms of action and side effect analyses. Additionally, effort in collecting and storing information on the constituents of the herbal plants and their chemical structure in public databases has expedited the process of analysing traditional medicine compounds. Examples of some of these databases can be seen in Table 1, although it should be noted that this list is not extensive.

In the next sections, the utilisation of these databases *in silico* to predict and understand the toxicity of TCM herbal medicine will be reviewed. There are three different methods covered in this mini review used for *in silico* toxicity studies of TCM herbal medicines, which are: 1) machine learning, 2) network toxicology, and 3) molecular docking. It should be noted that this review only covers the intrinsic toxicity of the compounds but not their extrinsic toxicity, where cultivation, processing, storage and distribution may contaminate the herbal raw material.

Table 1

List of several databases that are available for *in silico* toxicity studies of TCM herbal medicine. Note that this list is not exhaustive.

Database	Description	Notes	URL	Reference
<i>Compound-target database</i>				
ChEMBL	Currently in its 31st version, ChEMBL is a manually curated bioactivity database that contains 2,331,700 compounds and 15,072 targets.	These databases contained bioactivity data points of synthetic and natural compounds. Given the scarcity of bioactivity databases of TCM compounds, ChEMBL and PubChem are useful resources to build machine learning models.	https://www.ebi.ac.uk/chembl	[5]
PubChem	PubChem is an open chemistry database that currently contains 112 million compounds and 185,291 proteins.		https://pubchem.ncbi.nlm.nih.gov	[6]
<i>Protein structure database</i>				
RCSB PDB	The US data center for the global Protein Data Bank (PDB) archive of 3D structure data for proteins, DNA, and RNA molecules	Crystal structures of proteins can be obtained here for docking purposes. However, there is a limited number of membrane proteins available, which may restrict the toxicity study of TCM compounds	https://www.rcsb.org/	[7]
<i>Toxicity-related database</i>				
CTD	CTD (Comparative Toxicogenomics Database) is a manually curated database of chemical–gene/protein interactions, chemical–disease and gene–disease relationships.	These databases provide information on protein targets and their associated toxicity/disease outcome. In network toxicology, this information can be used for screening of toxic TCM compounds and explaining the mechanism of toxicity through further KEGG pathway analysis and GO annotation.	http://ctdbase.org/	[8]
TTD	Therapeutic Target Database (TTD) provides information about the known and explored therapeutic protein and nucleic acid targets, the targeted disease, pathway information and the corresponding drugs directed at each of these targets.		http://db.idrblab.net/ttd/	[9]
OMIM	Online Mendelian Inheritance in Man (OMIM), is a comprehensive, authoritative compendium of human genes and genetic phenotypes. It contains information on all known mendelian disorders and over 16,000 genes.		https://www.omim.org	[10]
<i>TCM-related database</i>				
SymMap	This database contains 698 TCM herbal preparation, 25, 795 ingredients and 20,965 targets of the ingredients	In comparison with other compound databases, the number of herbs contained are small and limited to those registered in the Chinese Pharmacopoeia. Hence, additional data on natural compounds for machine learning models needs to be obtained from other resources e.g ChEMBL	http://www.symmap.org/	[11]
TCMSP	TCMSP (TCM Systems Pharmacology) database contains 499 Chinese herbs registered in the Chinese pharmacopoeia with 29,384 ingredients, 3311 targets and 837 associated diseases.		https://tcmsp-e.com/tcmstp.php	[12]

Machine learning models

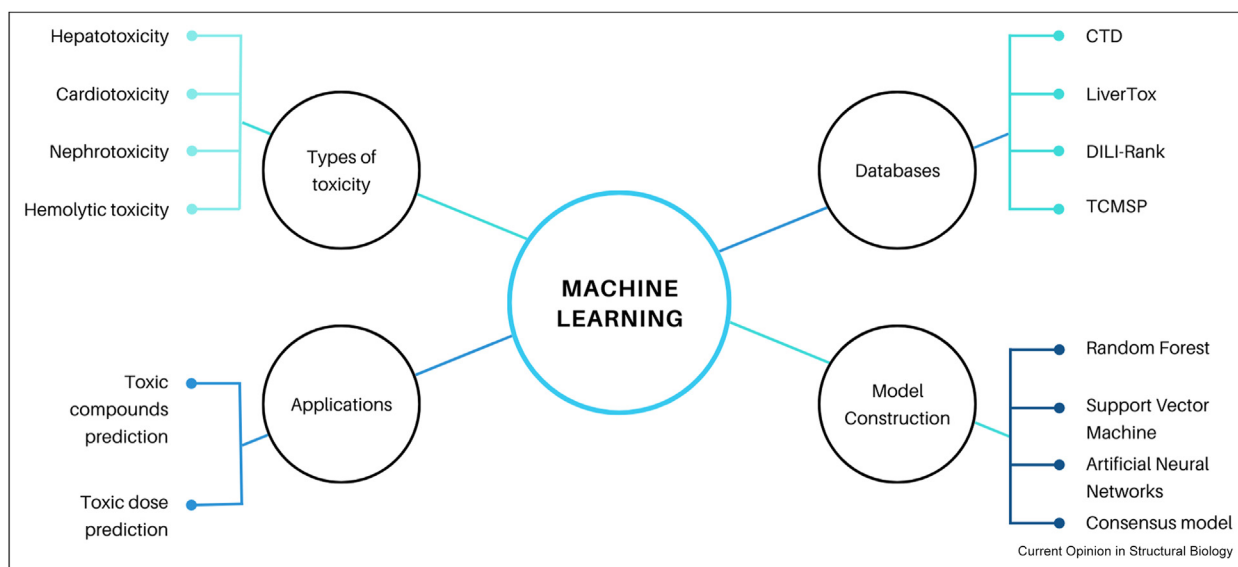
Machine learning is a subfield of artificial intelligence, which employs algorithms to simulate how humans learn to solve a problem or complete a task. A training set is used to train a machine learning model, where in toxicity prediction, high quality dataset consisting of either compound-target or compound-toxicity are usually used (see Figure 1). In the former, the targets are usually those that have been established to be associated with particular toxicity. Information on the compounds is usually related to their structures, but it can also contain their physicochemical properties. A ligand-based model is built when only the chemical structures are transformed into descriptors, and a proteochemometric model is built when both the chemical and protein structures are transformed. A machine learning algorithm will then functionally map the descriptors to the toxicity of interest. The algorithm can either be classification- or regression-based where the former predicts a discrete class label and the latter predicts a continuous value. Examples of machine learning algorithms that are usually used include k-Nearest Neighbour (kNN), Artificial Neural Network (ANN), Random Forest (RF), Naïve Bayes (NB). For more details on the different machine learning available, readers are directed to Ref. [13]. The process then proceeds to validation where models will be subjected to internal and external validations.

When looking at the application of machine learning in predicting TCM toxicities published since 2015, the majority of the studies involved hepatotoxicity

predictions [14–20]. This is likely due to hepatotoxic data being more abundant in public databases compared to other types of toxicity. Nonetheless, several other toxicities have been covered, which include cardiotoxicity [21], hemolytic toxicity [22], and nephrotoxicity [23]. The majority of these models are built using data from databases such as CTD [8], LivTox [24], and DILI-Rank [25], and these databases contain mostly synthetic compounds. TCM formulation contains solely natural compounds, and both types of compounds occupy different chemical spaces and have distinct properties. TCM-related databases such as TCMSP [12] were employed but their use is not as common as non-TCM-specific databases in building toxicity prediction models.

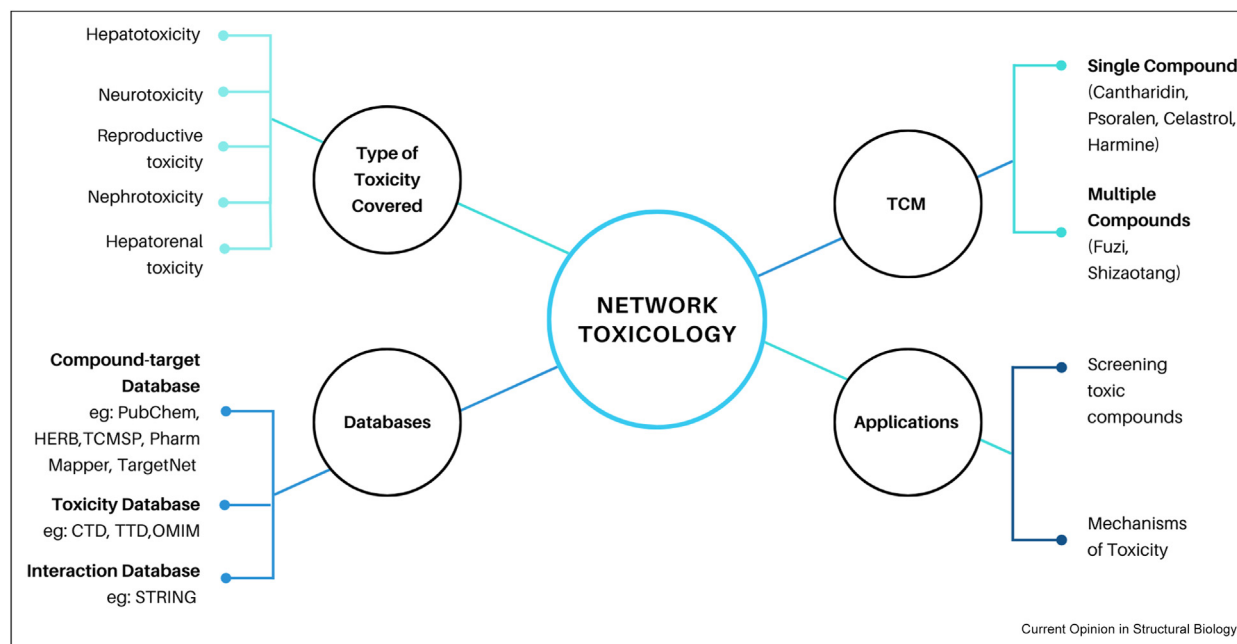
The majority of TCM toxicity prediction models built are classification-based, employing algorithms such as Random Forest [14,17,20,22], SVM [15,16,20] and Artificial Neural Network (ANN) [17,20]. Some studies employed multiple models, algorithms and descriptors to produce more robust consensus models to predict toxicity of TCM compounds [14,17]. He et al. [14], employed NaïveBayes, LibSVM, IBK, KStar, AdaboostM1-J48, Bagging-IBK, J48, and Random Forest algorithms to predict *Polygonum multiflorum* hepatotoxicity resulting in a balanced accuracy between 57.7% and 76.3% in the external validation. Those models were then combined using Naïve Bayes classifier to produce the final prediction where the combined classifier performed better with balanced accuracy of 78.2%. Similar results were observed by Wu et al. [17], when

Figure 1



Overview of TCM toxicity studies using machine learning approaches. The majority of studies involving machine learning in TCM toxicity prediction cover hepatotoxicity due to the availability of databases such as DILI-Rank that stores hepatotoxic data points.

Figure 2



Overview of TCM toxicity studies network toxicology. Information from various types of databases such as compound-target, toxicity and protein–protein interaction databases were used to construct a network to predict the toxic compounds in a TCM formulation and establish mechanisms of toxicity of either single or multiple compounds.

consensus prediction model was used to build a predictive model for hepatotoxicity. The consensus model (78%) showed higher Matthews Correlation Coefficient (MCC) than the best single classifier, MACCS-ANN (70%). The consensus model was used to screen the TCMSP database, resulting in the identification of *Bupleurum L. (Chaihu)* as the herb containing the highest number of hepatotoxic compounds.

Approaching toxicity prediction differently, Li et al. [15], utilised machine learning models to predict toxicity dose instead. Rather than using chemical structures as molecular descriptors, this study utilised *in vitro* hepatotoxicity data such as AUC and $C_{\max}$ obtained from experimental studies. Through reverse modeling, dosing schedules corresponding to probability of hepatotoxicity were predicted for emodin, matrine, triplidine and oxymatrine [15].

Although deep learning is fast gaining momentum in drug discovery, its application in TCM toxicity studies is still limited. To date, only one deep learning model has been reported. Chen et al. [20], compared the performance of a deep learning model called deep belief network (DBN) in screening potential hepatotoxic TCM-Western medicine combinations. It was found that DBN outperformed nine other machine learning algorithms with classification accuracy (CA) of 82.2% compared to the rest with a CA of 67.5–76.7%.

Network toxicology

Network toxicology is another *in silico* method used to predict toxicity of TCM compounds. Network toxicology examines how a drug or combination of drugs systematically perturbs multiple targets rather than single targets that cause particular toxicity (see Figure 2) [26]. This approach shares a similar holistic philosophy to TCM, where the synergistic action of multiple active phytochemicals of the herbs act on multiple targets/pathways [27].

In constructing network toxicology for TCM compounds, compound-target nodes are first constructed for a compound by either retrieving from public databases such as PubChem [6], HERB [28], TCMSP [12], BATMAN-TCM [29] and/or subjected to target prediction (e.g. Pharm Mapper [30], Swiss Target Prediction [31], TargetNet [32], DRAR-CPI [33]) The target-toxicity nodes are then constructed by retrieving the information from toxicity databases such as CTD [8], TTD [9] or OMIM [10] (see Table 1). KEGG pathway enrichment analysis and Gene Ontology (GO) annotation are employed to describe the characteristics of hub genes. Next, interactions between the protein nodes are established through methods such as phylogenetic profiling [34] and protein–protein interaction (PPI) [35] using STRING [36]. A network connecting all possible nodes is then constructed using methods such as topological analysis, which is the most often

employed. Topological analysis establishes the best degree of connectivity between nodes and edges through centrality analysis, clustering, shortest paths, or motif search. The network can then be visualized using software such as Cytoscape [37] and Pajek [38].

In the last five years, the majority of network toxicology for TCM toxicity prediction covers hepatotoxicity [39,40], neurotoxicity [41,42], reproductive toxicity [43], nephrotoxicity [44], and cardiotoxicity [45]. In addition, network toxicology is either done on a single compound or multiple compounds of a TCM formulation. In the case of single compound, network toxicology is usually carried out after *in vitro* or *in vivo* experiments. Studies by He et al. [44], and Sun et al. [39], performed UHPLC-QE/MS-based metabolomics on Cantharidin and DARTS-based proteomics on Psoralen, respectively prior to network toxicology. He et al. [44], identified 132 metabolic targets from metabolomics analysis and additional 38 predicted targets. Meanwhile, Sun et al. [39], determined seven proteins through DARTS-based proteomics, 106 proteins predicted targets and 45 targets from literature survey. Network toxicology was constructed using these proteins as explained previously. The network showed that Cantharidin caused nephrotoxicity by inhibiting the glycerophospholipid and sphingolipid pathways, phosphatidylethanolamine, phosphatidylcholine, MAPK3, and PLD2 [44]. Meanwhile, Psoralen induced hepatotoxicity by binding to ABL1, and regulated Nrf2, which increased the level of ROS, and affected the expression of mTOR, leading to cell death [39]. Sun et al. [46], Liu et al. [45], and Sun et al. [41], first confirmed the toxicity of Terrestrosin D (TED), Celestrol and Harmine, respectively *in vivo*, followed by network toxicology. Network toxicology showed that the main targets that played an important role in hepatorenal toxicity induced by TED were HSP90AA1, CNR1, DRD2 through multiple pathways (cAMP signaling pathway, pathway in cancer and G-protein coupled receptor signaling pathway) [41]. Moreover, the ten main targets that induced cardiotoxicity by Celestrol were INS, GAPDH, TNF, VEGFA, CASP3, IL10, MMP9 and MAPK8 through pathways such as TNF, NOD-like receptor and Prolactin signaling pathways [45]. In the case of Harmine, 15 targets were predicted for its neurotoxicity where one of them is acetylcholinesterase. Harmine could directly inhibit the activity of acetylcholinesterase which causes excessive accumulation of acetylcholine.

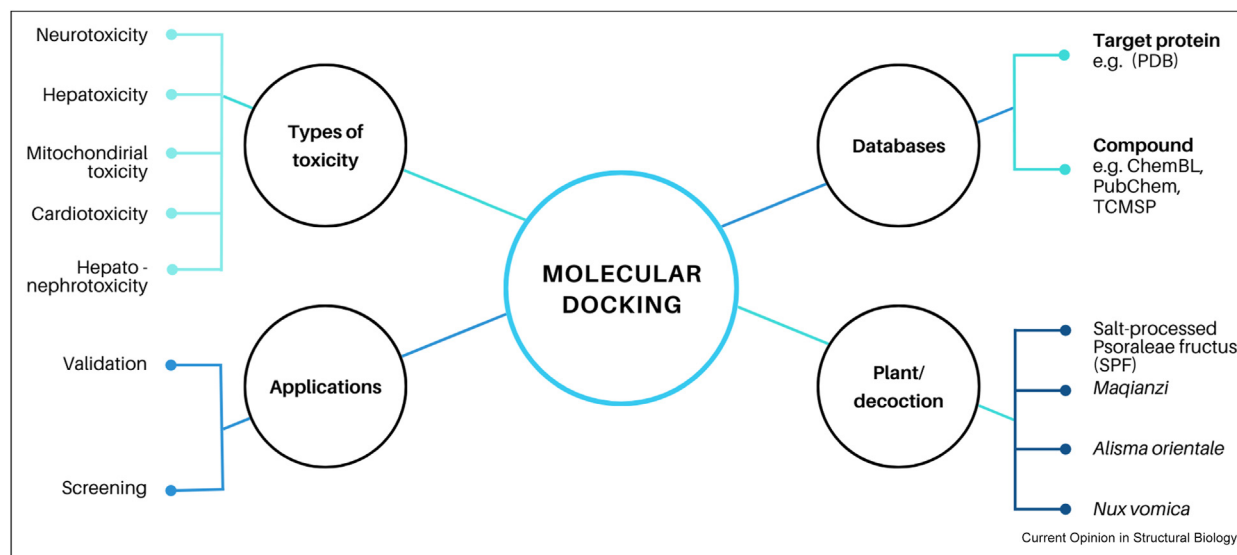
When multiple compounds are being analysed, network toxicology is performed first before experimental validation. Network toxicology first screens which compound(s) in the formulation are toxic by looking at the targets mapped to the compounds or targets predicted. Compounds where the associated targets are linked to specific toxicity will further undergo PPI analysis, KEGG pathway enrichment analysis and GO annotation.

In a study conducted by An et al. [42], network toxicology first screened three key toxic compounds (uracil, palmitic acid, aconitine) in *Fuzi* that induced neurotoxicity. These compounds induced neurotoxicity by ten core targets (ALB, AKT1, TP53, IL6, TNF, INS, CASP3, IL1B, EGFR and BDNF) through pathways such as pathways of neurodegenerative multiple disease, MAPK signaling pathway, HIF-1 signaling pathway, TNF signaling pathway, and various infection and tumor-specific pathways. Then, molecular docking and *in vitro* studies were conducted to validate the mechanism of neurotoxicity involving aconitine in *Fuzi*. Zhang et al. [40], performed serum pharmacochimistry and network toxicology to screen potential toxic components of Heishunpian (HSP), a processed product of *Fuzi*. 22 components of HSP were determined to induce hepatotoxicity by regulating targets such as AKT1, IL2, F2, GSR, and EGFR through multiple pathways (Th17 cell differentiation, Jak-STAT signaling pathway, and glutathione metabolism). In addition, Zhang et al. [47], conducted UFLC-Q-TOF-MS analysis to identify 42 compounds in Shizaotang (SZT) and vinegar Shizaotang (VSZT), followed by network toxicology to look into both effects and toxicity of the formulation. In network toxicology analysis, the top 3 toxic compounds of SZT/VSZT were 3-O-(2,3-dimethylbutanoyl)-13-O-dodecanoyl-20-deoxyingenol, Kansuinin E and (3 β)-3-hydroxyanosta-8,24-diene-7,11-dione. Meanwhile, the top 3 targets associated with their toxicity were MAPK3, MAPK1, and AKT1, which affect pathways such as PI3K-Akt signaling pathway, and Epstein-Barr virus infection. Then, these two studies were validated *in vivo*, where the compounds were administered in rats followed by histopathological examination [47].

Molecular docking

Molecular docking establishes the interaction between a small compound and a large biological molecule e.g. protein, DNA, RNA at the atomic level (see Figure 3). The structure of the compound can be obtained from databases such as ChEMBL [5] and PubChem [6], whereas the structure of the biological molecule is retrieved from Protein Data Bank (PDB) [7]. In the event where no crystal structures are available, homology modeling can be used to obtain the three-dimensional structure of proteins. There are two stages in docking, which are conformations sampling and scoring. The sampling process searches the conformational space described by the free energy landscape, where the energy is approximated by the scoring function. Here, the scoring function should be able to associate the native bound-conformation to the global minimum of the energy hypersurface. Various molecular docking software e.g. AutoDock [48] and Glide [49,49] and web servers e.g. PatchDock [50] and ClusPro [51] that use different algorithms have been developed to perform the molecular docking exercise. The binding conformation of the protein-ligand complex can be

Figure 3



Overview of TCM toxicity studies using molecular docking. There are several types of studies that apply docking, where it is mainly used as a validation for results produced by network toxicology, *in vitro* and *in vivo* studies.

visualized using 3D visualizer software such as PyMOL [52].

Based on literature published within the past 5 years, molecular docking has been used to predict different types of toxicity of TCM compounds such as hepatotoxicity [53,54], neurotoxicity [55], hepato-nephrotoxicity [56], mitochondrial toxicity [57] and cardiotoxicity [58]. Unlike machine learning, the application of docking covers more types of toxicity as it is not dependent on a substantial amount of bioactivity data. However, its application is dependent on the identification of a protein target(s) that is known and validated to cause certain toxicity. Similar to machine learning, docking focuses on a single compound, and not in combination as in network toxicology. Docking is advantageous to natural products e.g TCM compounds especially when docked against a shallow binding site. Here, the confidence level of natural products or large size molecules docked is higher when compared to small molecules.

Docking on its own is rarely employed for toxicity prediction. In combination with other *in silico* methods, docking is often used for validation and at times for screening purposes. Docking is usually performed to validate findings from network toxicology [55,56], *in vitro* [54] or *in vivo* [58] studies. In network toxicology, docking validates the compound-target node(s) of a network. A lower binding energy would further validate the potential compound-target interaction, which then narrows down and identifies the most likely pathway and biological process that is involved in the toxicity event. Dai et al.

[55], utilised such experimental protocol in determining strychnine-induced neurotoxicity pathway of a *Maqianzi* herb. Protein targets and central nervous system-related gene sets were first identified *via* network toxicology. Following this, three neurotoxic compounds ((*S*)-strychnine, isobrucine and stigmastrol) and CHRM1 were selected for the docking validation. The predicted ligands showed lower binding energy with protein CHRM1 than the co-crystallized ligands and formed hydrogen bond interaction not seen in the co-crystallized ligand, which confirmed their involvement in neurotoxicity pathway. Zhang et al. [54], performed *in vitro* HepG2 cytotoxic assay, and gene expression and western blot analysis of CYP1A2 in mice liver where activation of the enzyme can aggravate hepatotoxicity. These analyses were conducted to test *Psoralea Fructus* (PF) effect on the liver. Further docking exercise was performed to confirm aryl hydrocarbon receptor (AhR) interaction with psoralene and isopsoralene that may be involved in the hepatotoxicity mechanism.

Only a few studies used docking for screening purposes [53,57] to identify compounds that may be responsible for the toxicity of interest. Shen et al. [57], established a platform integrating dual *in vitro* assays and molecular docking, to predict mitochondrial toxicants of 35 TCM injections. From initial *in vitro* screening, a total of 622 ingredients were identified as mitotoxigants, and docking was performed to screen potential toxic compounds against NADH dehydrogenase (ubiquinone) flavoprotein 1, mitochondrial (NDUFV1). The docking result narrowed down the potential intoxicants to 87 compounds, with scutellarin, rutin, phillyrin, and baicalin

being the top candidates with strong affinity towards the protein. Further dual *in vitro* assays of these compounds confirmed the potential of mitochondrial intoxicants. Similarly, Wu et al. [53], screened the components of Salt-processed *Psoralea fructus* (SPF) that could cause hepatotoxicity using molecular docking. Further *in vivo* test was carried out, using SPF decoction at different concentrations and followed by several analyses e.g., liver injury and western blot analyses.

The 3D visualization in docking allows for the understanding of the interaction between ligand and protein, specifically the functional groups that are involved in the ligand–protein complex. Consequently, potential structural alerts can be generated as reported by Yuan et al., [59]. In this study, the effect of strychnine and brucine, the main ingredients of *Nux vomica* were observed on hERG channels which are commonly used to screen for compound cardiac toxicity. The docking result suggested that the presence of an oxygen to compensate for the N⁺ charge of the compound, could be a useful strategy for reducing hERG potency and increasing the safety margin of alkaloid-type compounds in drug development [59]. Furthermore, the presence of this functional group in alkaloids could be used as an alert on potential toxicity of future compounds.

Conclusion

In this review, three different *in silico* methods to study the toxicity of TCM herbal medicines were presented, which are machine learning, network toxicology and molecular docking. For each method, its application and implementation e.g., single classifier *vs.* multiple classifier, single compound *vs.* multiple compounds, validation *vs.* screening, were reviewed. Overall, *in silico* methods provide a data-driven prediction of potential toxicity of TCM formulation that has been validated *in vitro* and/or *in vivo*. However, it is still limited to several toxicities. Additionally, most of the studies are still limited to a single compound when TCM formulation involves multiple plants containing multiple compounds. To increase coverage of different toxicities, other methods such as text mining should be considered in addition to structural information. With the limited data points on TCM compounds and natural products, it may be rational to seek data from government agencies, news and/or social media sites and perform as was done during the COVID-19 pandemic. This may help increase the type of toxicity covered, as well as speed up the progress in this field. However, data from these sources need to be validated first. In the future, more studies should be done to analyse and test the combination of compounds on the front end i.e., to generate data for *in silico* modeling and back end i.e., validate findings from prediction models, of the process.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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- * of special interest
- ** of outstanding interest

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List of abbreviations

TCM:	Traditional Chinese Medicine
SVM:	Support Vector Machine
ANN:	Artificial Neural Network
MCC:	Matthew Correlation Coefficient
DBN:	Deep Belief Network
CA:	Classification Accuracy
SGD:	Stochastic Gradient Decent
kNN:	K-Nearest Neighbor
DT:	Decision Tree
RF:	Random Forest
LR:	Logistic Regression
NB:	Naive Bayes
HERB:	High-Troughput Experiment- And Reference-Guided Database
TCMSP:	Traditional Chinese Medicine Systems Pharmacology
BATMAN-TCM:	Bioinformatics Analysis Tool For Molecular Mechanism Of Traditional
DR4R-CPI:	Drug Repositioning And Adverse Reactions Via The Chemical-Protein Interactome
CTD:	Comparative Toxicogenomics Database
TTD:	Therapeutic Target Database
OMIM:	Online Mendelian Inheritance In Man
KEGG:	Kyoto Encyclopedia Of Genes And Genomes
GO:	Gene Ontology
PPI:	Protein–Protein Interactions
STRING:	Search Tool For The Retrieval Of Interacting Genes/Proteins
DARTS:	Drug Affinity Responsive Target Stability
UHPLC-QE/MS:	Ultra-High Performance Liquid Chromatography-Quadrupole Time-Of-Flight Mass Spectrometry
UFLC-Q-TOF-MS:	Ultra-Fast Liquid Chromatography/Quadrupole-Time Of Flight Tandem Mass Spectrometry
MAPK3:	Mitogen-Activated Protein Kinase 3
PLD2:	Phospholipase D2
ABL1:	Albumin 1
Nrf2:	Nuclear Factor Erythroid 2-Related Factor 2
ROS:	Reactive Oxygen Species
mTOR:	Mammalian Target Of Rapamycin
HSP90AA1:	Heat Shock Protein 90a
CNR1:	Cannabinoid Receptor 1
DRD2:	Dopamine D2 Receptor
cAMP:	Cyclic AMP
INS:	Insulin
GAPDH:	Glyceraldehyde 3-Phosphate Dehydrogenase
TNF:	Tumor Necrosis Factor
VEGFA:	Vascular Endothelial Growth Factor A
CASP3:	Cysteine-Aspartic Acid Protease
IL10:	Interleukin 10
MMP9:	Matrix Metalloproteinase 9
MAPK8:	Mitogen-Activated Protein Kinase 8
NOD-like:	Nucleotide-Binding Oligomerization Domain-Like
ALB:	Albumin
AKT1:	Akt Serine/Threonine Kinase 1
TP53:	Tumor Protein P53
IL6:	Interleukin 6
IL1B:	Interleukin-1 Beta
EGFR:	Epidermal Growth Factor Receptor
BDNF:	Brain-Derived Neurotrophic Factor
IL2:	Interleukin 2
F2:	Coagulation Factor 2
GSR:	Glutathione Reductase
Th17:	T Helper 17 Cells
Jak-STAT:	Janus Kinase-Signal Transducer And Activator Of Transcription
MAPK1:	Mitogen-Activated Protein Kinase 1
PI3K-Akt:	Phosphatidylinositol 3-Kinase (Pi3k)/Protein Kinase B
PDB:	Protein Data Bank
DNA:	Deoxyribonucleic Acid
RNA:	Ribonucleic Acid
CHRM1:	Muscarinic Acetylcholine Receptor M1
AhR:	Aryl Hydrocarbon Receptor
SPF:	Salt-Processed Psoralea Fructus
NDUFV1:	NADH Dehydrogenase (Ubiquinone) Flavoprotein 1, Mitochondrial
CYP1A2:	Cytochrome P450 1A2
hERG:	Human Ether-A-Go-Go Related Gene